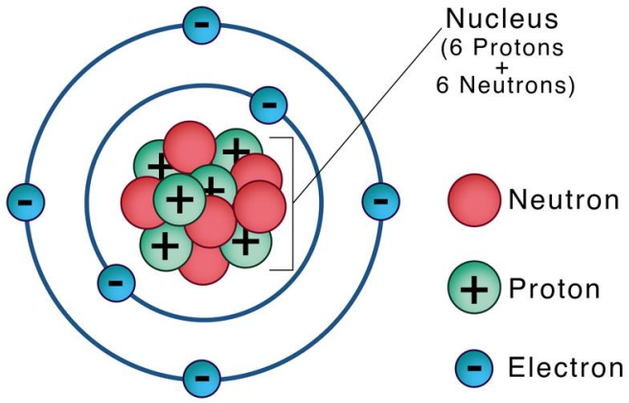
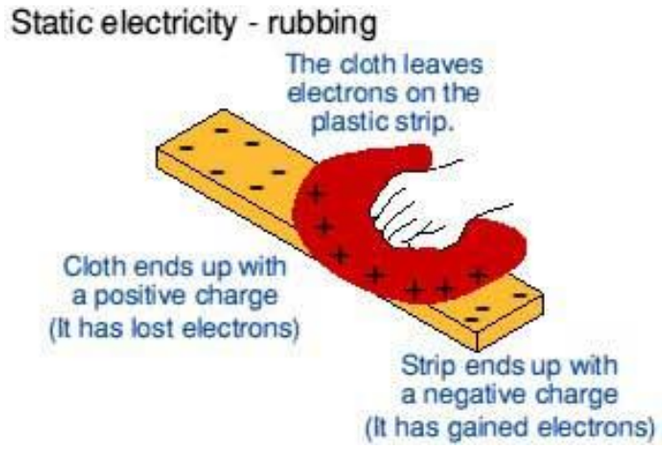
Knowledge Organiser
Year 7: Electricity – Charge and Current

Why are some objects charged?
- All objects are made up of atoms, which have a **nucleus** containing **protons** and **neutrons**, and **electrons** orbiting around the nucleus
- Electrons have a **negative charge**, protons have a **positive charge** and neutrons have **no charge**
- The charge of an object is determined by the number of electrons and protons it has
- An object that has more electrons than protons is negatively charged, while an object with more protons than electrons is positively charged



This atom has **no overall charge** because it has the same number of positive protons and negative electrons

What is static electricity?
- **Static electricity** is the build-up of electrical charge on the surface of an object
- This can happen when two objects are rubbed together**, transferring electrons** between them
- Objects with **opposite charges** (positive and negative) will be attracted to each other
- Objects with **the same charge** (positive and positive or negative and negative) will repel each other

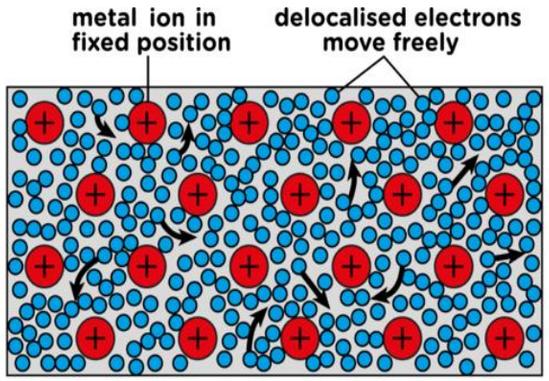


A diagram showing how static electricity occurs when rubbing a plastic rod with a cloth

Conductors and Insulators
- Conductors are materials that allow electrical current to flow through them easily
- Metals are very good conductors of electricity
- Insulators are materials that do not allow electrical current to flow through them easily
- Rubber is a good insulator and a bad conductor

Material	Conductor or Insulator?
Metal	Conductor
Rubber	Insulator
Plastic	Insulator
Wood	Insulator
Salt water	Conductor
Air	Insulator

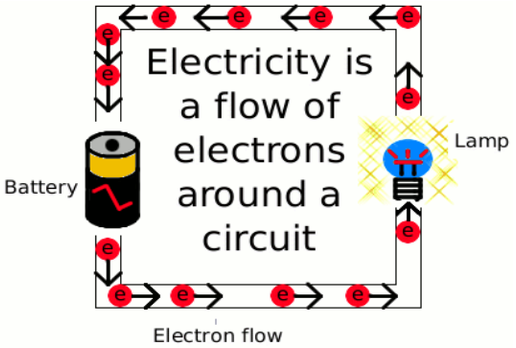
A table listing common conducting and insulating materials



Metals are good conductors because they have lots of **free electrons.**

These can move and carry energy.

Electrical current?
- Electrical current is the flow of electric charge around a circuit
- Usually this is when electrons flow through wires
- A complete circuit is needed for electrical current to flow
- The charged particles carry energy from a power source and transfer it to components in the circuit.



A diagram showing how electrical current flows through a circuit, delivering energy to a bulb.

Keyword	Definition
Charge	Measure of electrical energy
Conductor	Material that allows electrical current to flow through easily
Insulator	Material that does not allow electrical current to flow through easily
Current	The flow of electric charge around a circuit
Electron	Negative subatomic particle which orbits the nucleus
Static	Build up of electric charge caused by friction

RAG Yourself!

Learning Outcomes;	RAG
Describe how charged particles interact	
Describe what causes static electricity	
Draw a labelled diagram of atomic structure	
State which particle moves in an electrical current	
Compare electrical conductors and insulators	
Use a simple model, such as flowing water, to explain how electrical current transfers energy	

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Conductors and Insulators

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